Gunjan Joshi, Ph.D

Machine-Learning Scientist, Helmholtz Zentrum Dresden Rossendorf, Germany

EDUCATION

Oct 2021 – Sept 2024
Doctor of Philosophy in Electrical Engineering and Information Systems, The University of Tokyo, Japan
Degree awarded: 20 September 2024
Advisor: Prof. Akira Hirose and Associate Prof. Ryo Natsuaki (Co-Advisor)
Thesis title: *Inverse Mapping for Neural Network Explainability and its Applications in Earth Observation*
Focus Areas: Explainable Machine Learning, Remote Sensing, Multimodal Sensor Data Fusion, Cryosphere
Minors: *Global Future Faculty Development Program* (Center for Research and Development of Higher Education); *Global Leader Program for Social Design and Management* (Graduate School of Public Policy); *Science, Technology and Innovation Governance Education Program* (Graduate School of Public Policy), The University of Tokyo

Oct 2019 – Sept 2021
Master of Engineering in Electrical Engineering and Information Systems, The University of Tokyo, Japan
Advisors: Prof. Akira Hirose and Associate Prof. Ryo Natsuaki
Thesis title: *Analysis of Significant Factors in Multi-Sensor Satellite Data Fusion for Earthquake Damage Assessment*

Sept 2012 – July 2016
Bachelor of Engineering in Electronics & Communication Engineering, Visvesvaraya Technological University, India
Thesis title: *FPGA Implementation of Channel Emulator for Testing of Wireless Air Interface using VHDL*

PROFESSIONAL EXPERIENCE

Oct 2024 – Present
Machine-Learning Scientist, Helmholtz-Zentrum Dresden-Rossendorf, Germany
Supervisors: Dr. Peter Steinbach *(Department of Computational Science)* & Prof. Dr. Pedram Ghamisi *(Department of Exploration)*
Project: The 3D-ABC Foundation Model: Towards Global 3D Above and Below Ground Carbon Stocks [Helmholtz Foundation Model Initiative]

Feb 2023 – March 2023
Guest Scientist, German Aerospace Center (DLR), Germany
Supervisor: Dr. Andreas J. Dietz *(Team Polar and Cold regions, German Remote Sensing Data Center)*
Project: Multi-sensor Multi-temporal Mapping of Swiss Alpine Glaciers using Explainable AI Techniques

April 2018 – Sept 2019
Research Scholar, Electrical Engineering and Information Systems, The University of Tokyo, Japan

Dec 2016 – March 2018
RF Engineer, TATA Advanced Systems, Bangalore, India

July 2016 – Dec 2016
Network Engineer, Ericsson Global, Bangalore, India

RECOGNITIONS

Honors

Oct 2025

Nov 2023 **Rising Stars Women in Engineering** – Selected participant for The University of Tokyo

Awards

July 2023 **Best Poster Award** from Department of Electrical Engineering and Information Sciences, The University of Tokyo.

July 2022 **First Prize Three Minutes Thesis (3MT) Competition**, IEEE International Geoscience and Remote Sensing Symposium

Nov 2021 **Japan Society of Photogrammetry and Remote Sensing (JSPRS) Award**, Asian Conference on Remote Sensing

Grants

July 2023 **Remote Sensing Technology Center of Japan (RESTEC) Research Grant 2023**, *PI: Gunjan Joshi*, 1 million JPY

IEEE-GRSS IDEA Professional Development Microgrants

Feb 2023 **The University of Tokyo Overseas Research Training (Musha Shugyo) Grant**

Scholarship

April 2018 – Sept 2024 **Japanese Government (MEXT) Monbukagakusho** Scholarship

TEACHING AND TRAINING ACTIVITIES

April 2025 **Course Instructor** for *Introduction to Machine Learning*, HIDA (Helmholtz Information and Data Science Academy)

Oct 2025 **Course Instructor** for *Introduction to Deep Learning*, HIDA (Helmholtz Information and Data Science Academy)

Nov 2025 **Trainer** for the *HPC-Gateway Fall School*, delivering sessions on Machine Learning

Talk on *"Use of Modern Large Language Models to Support Your Scientific Writing"* for the HZDR PhD Students' Writing Sprint

2020–2024 **Teaching Assistant** for *Advanced Academic Writing and Presentation*, School of Engineering, The University of Tokyo

Academic Writing Consultant for *English Writing Center*, School of Engineering, The University of Tokyo

INVITED TALKS

Jan 2026 **Invited Talk**, International Institute of Information Technology Bangalore (IIIT-B), Bengaluru, India. *"Opening the Black Box: Understanding Multi-Sensor Fusion through Explainable AI"*.

Dec 2025 **Invited Speaker** for the IEEE GRSS Young Professionals Summit 2025, Colombo, Sri Lanka, to deliver talk on *"Advancing Earth Observation with Foundation Models"*.

Oct 2025 **Invited Speaker** for the IEEE GRSS Sri Lanka Chapter, delivering the talk *"Making Sense of Sensors: Using Explainable AI for Multi-Sensor Data Fusion"*.

Aug 2025 **Invited Speaker** in the IGARSS 2025 tutorial *"Machine Learning-Based Integration of Optical and SAR Data for Monitoring Earth's Dynamic Processes"*, talk titled *"Explainable AI and Its Applications in Multi-Sensor Earth Observation"*.

Dec 2024 **Invited Speaker** at the University of Melbourne, Australia for lecture on *"Explainable AI and Its Applications in Earth Observation"*.

ACADEMIC SERVICES

March 2025 – Present	**Editorial Board - Column Editor**, IEEE Geoscience and Remote Sensing Magazine
July 2024 – Present	**Webinar Strategist, Digital Content Team Lead**, IEEE Geoscience and Remote Sensing Society
March 2023 – July 2024	**Social Media Ambassador**, IEEE Geoscience and Remote Sensing Society

REVIEWER

1 IEEE Open Journal of the Computer Society
2 IEEE Geoscience and Remote Sensing Magazine
3 European Journal of Remote Sensing
4 IEEE Transactions on Geoscience and Remote Sensing
5 Helmholtz AI Conference 2025
6 Elsevier Advances in Space Research
7 Asia-Pacific Conference on Synthetic Aperture Radar 2023
8 IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

PUBLICATIONS

Overall citation metrics

Total citations: 67, **h-index:** 5; **i10-index:** 4.

(Date accessed 02 Feb 2025)

Peer-reviewed Journal Publications

1 J. Steiner, W. Armstrong, W. Kochtitzky, R. McNabb, R. Aguayo, T. Bolch, F. Maussion, V. Agarwal, I. Barr, N. R. Baurley, M. Cloutier, K. DeWater, F. Donachie, Y. Drocourt, S. Garg, **G. Joshi***, B. Guzman, S. Kutuzov, T. Loriaux, C. Mathias, B. Menounos, E. Miles, A. Osika, K. Potter, A. Racoviteanu, B. Rick, M. Sterner, G. D. Tallentire, L. Tielidze, R. White, K. Wu, and W. Zheng, "Global mapping of lake-terminating glaciers," in *Earth System Science Data* (preprint, in review), 2025, doi: 10.5194/essd-2025-315. *Journal Impact factor (2024): 11.6, ; 5-year Impact Factor : 13.9*

2 **G. Joshi***, C. A. Baumhoer, A. J. Dietz, R. Natsuaki and A. Hirose, "Multisensor Glacier Surface Classification Using Confidence-Aware Explainable Inverse-Mapping Neural Network," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024, doi: 10.1109/JSTARS.2024.3454789. *Journal Impact factor (2024): 5.3*

3 **G. Joshi***, R. Natsuaki and A. Hirose, "Application of Inverse Mapping for Automated Determination of Normalized Indices Useful for Land Surface Classification," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2023, doi: 10.1109/JSTARS.2023.3308049. *Journal Impact factor (2024): 5.3, Citations: 10*

4 **G. Joshi***, R. Natsuaki and A. Hirose, "Neural-network fusion processing and inverse mapping to combine multi-sensor satellite data and analyze the prominent features," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2023, doi: 10.1109/JSTARS.2023.3247788. *Journal Impact factor (2024): 5.3, Citations: 16*

Peer-reviewed International Conferences

5 G. Grosse, J. Hashemi, L. van Delden, T. Lübker, I. Nitze, J. Strauss, S. Kruse, P. Ghamisi, P. Steinbach, A. Rizaldy, **G. Joshi***, W. Yu, R. Gloaguen, G. Cavallaro, E. Zandi, R. Sedona, S. Hashim, S. Mandal, M. Herold, Q. Song, S. Besnard, M. Urbazaev, M. Sips, A. Huth, I. Hajnsek, and M. Pardini, "3D-ABC: A Foundation Model for Global Terrestrial 3D Above and Below Ground Carbon Stock Mapping," *IGARSS 2025 – 2025 IEEE International Geoscience and Remote Sensing Symposium*, Brisbane, Australia, 2025, pp. 862–866, doi: 10.1109/IGARSS55030.2025.11242519.

6 J. Hashemi, S. Besnard, G. Cavallaro, P. Ghamisi, R. Gloaguen, I. Hajnsek, S. Hashim, M. Herold, U. Herzschuh, A. Huth, **G. Joshi***, S. Kruse, T. Lübker, S. Mandal, I. Nitze, M. Pardini, A. Rizaldy, F. Röseler, R. Sedona, M. Sips, Q. Song, P. Steinbach, J. Strauss, M. Urbazaev, L. van Delden, W. Yu, E. Zandi, and G. Grosse, "Towards Enhancing Permafrost Carbon Stock Estimates using 3D-ABC: a Multimodal Foundation Model for Geospatial Analytics," extended abstract, submitted to *IGARSS 2025 – 2025 IEEE International Geoscience and Remote Sensing Symposium*, 2025.

7 G. Grosse, P. Ghamisi, G. Cavallaro, M. Herold, A. Huth, I. Hajnsek, and **the 3D-ABC Team***. "Towards a Foundation Model for Global Terrestrial 3D Above and Below Ground Carbon Stock Mapping (3D-ABC)," *European Geosciences Union (EGU) General Assembly*, Vienna, Austria, April 2025. [EGU25-19378]

8 **G. Joshi***, R. Natsuaki, and A. Hirose, "Complex-Valued Neural Network Inverse Mapping for Explainability in PolSAR/InSAR Applications," *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium*, Athens, Greece, 2024, pp. 2000–2003, doi: 10.1109/IGARSS53475.2024.10640643.

9 **G. Joshi***, C. A. Baumhoer, A. J. Dietz, R. Natsuaki, and A. Hirose. "Estimating Snow Line Altitude by Optical and SAR Data Fusion: Inverse-mapping Explainable Neural Network-Based Approach - Case Study of the Great Aletsch Glacier" *15th European Conference on Synthetic Aperture Radar (EUSAR 2024)*, Munich, Germany, 2024, pp 399-404. doi: https://ieeexplore.ieee.org/document/10659607.

10 **G. Joshi***, R. Natsuaki and A. Hirose, "Fusion of ALOS-2/PALSAR-2 and Sentinel-1 Polarimetric Imagery with Explainable Neural Network-based Confidence Level Estimation for Glacier Surface Classification," 2023 *8th Asia-Pacific Conference on Synthetic Aperture Radar (APSAR)*, Bali, Indonesia, 2023, pp. 1-6, doi: 10.1109/APSAR58496.2023.10388723. *Citations: 1*

11 **G. Joshi***, R. Natsuaki and A. Hirose, "Automated determination of Normalized Indices useful for Glacier Surface Classification," *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium, Pasadena, CA, USA, 2023*, pp. 215-218, doi: 10.1109/IGARSS52108.2023.10281784. *Citations: 1*

12 **G. Joshi***, R. Natsuaki and A. Hirose, "Neural Network Model for Multi-Sensor Fusion and Inverse Mapping Dynamics for the Analysis of Significant Factors," *IGARSS 2022 - 2022 IEEE International Geoscience and Remote Sensing Symposium, 2022*, pp. 473-476, doi: 10.1109/IGARSS46834.2022.9884409. *Citations: 7*

13 **G. Joshi***, R. Natsuaki, and A. Hirose, "Multi-Sensor Satellite-Imaging Data Fusion for Earthquake Damage Assessment and the Significant Features," *Proc. - 2021 7th Asia-Pacific Conf. Synth. Aperture Radar, APSAR 2021*, 2021, doi: 10.1109/APSAR52370.2021.9688438. *Citations: 12*

14 **G. Joshi***, R. Natsuaki, and A. Hirose. "SAR and Optical Sensor Data Fusion for Earthquake Damage Assessment and Analysis of the Significant Features," The 42nd Asian Conference on Remote Sensing (ACRS 2021), Can Tho, Vietnam. [ACRS Proceedings].

15 **G. Joshi***, P. R. Prasad and A. Singh, "FPGA Implementation of Channel Emulator for Testing of Wireless Air Interface Using VHDL," *2016 IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, Bangalore, India, 2016, pp. 728–732, doi: 10.1109/RTEICT.2016.7807920. *Citations: 17*

Others

16 J. Hashemi, S. Besnard, G. Cavallaro, P. Ghamisi, R. Gloaguen, I. Hajnsek, S. Hashim, M. Herold, U. Herzschuh, A. Huth, **G. Joshi***, S. Kruse, T. Lübker, S. Mandal, I. Nitze, M. Pardini, A. Rizaldy, F. Röseler, R. Sedona, M. Sips, Q. Song, P. Steinbach, J. Strauss, M. Urbazaev, L. van Delden, W. Yu, E. Zandi, and G. Grosse. "Permafrost Soil Carbon Stock Estimation using a Multimodal Geospatial Foundation Model, 3D-ABC," *ESA–NASA International Workshop on AI Foundation Models for Earth Observation*, ESA-ESRIN, Frascati, Italy, 5–7 May 2025. [PDF]

17 A. Hirose and **G. Joshi***. "Advanced Earth Observations Using Explainable AI to Monitor Glaciers for Water Resource Evaluation," *IEEE Humanitarian Activities Workshop with AI Technologies, IEEE World Congress on Computational Intelligence (WCCI)*, 2024.

18 A. Hirose, R. Natsuaki, and **G. Joshi***. "Explainable Artificial Neural Networks in PolSAR Data Processing," *Joint PI Meeting of JAXA Earth Observation Missions FY2024*, 18–22 November 2024.

19 A. Hirose and **G. Joshi***. "Advanced Earth Observations Using Explainable AI to Monitor Glaciers for Water Resource Evaluation," *IEEE Humanitarian Activities Workshop with AI Technologies (IJCNN Workshop on WCCI2024)*, June 30 2024.

20 A. Hirose, R. Natsuaki, and **G. Joshi***. "Fusion of Synthetic-Aperture-Radar and Optical Data and Explainability in Neural Network Processing," *The Joint PI Meeting of JAXA Earth Observation Missions FY2023*, November 6 - 10, 2023.

21 **G. Joshi**, R. Natsuaki, and A. Hirose. "Satellite Data Fusion for Damage Assessment and Analysis of the Significant Features," *Institute of Electronics, Information and Communication Engineers Technical Committee on Space, Aeronautical and Navigational Electronics (IEICE-SANE)*, vol. 121, no. 154, SANE2021-29, pp. 37-42, August 2021 [IEICE Technical Report].